Anirudhhan Raghuraman

anirudhhan.raghuraman@gmail.com | Linked In: Anirudhhan Raghuraman | GitHub: Heatwave591 | Ph: (380)282-4118

EDUCATION

The Ohio State University **June 2026**
Master's in electrical and computer engineering, Specialization in Circuits *GPA: 3.5/4*
PES University **June 2024**
Bachelor's in electrical and communication engineering, Specialization in VLSI *GPA: 3.5/4*

SKILLS

Programming & Tools: C, Python, Verilog, FPGA, Xilinx, Quartus MATLAB, SPICE, KiCAD, Git, Shell, Cadence, Excel, LATEX, Assembly, Wireshark, GNS3, Raspbian, SQL, RTOS, Sensors, Communication protocols
Concepts: Routing, Computer and Embedded System Arch., VLSI Design, Design Verification, Embedded Firmware, PCB Design, Micro controllers, Digital Electronics, Computer Hardware, Internet of Things, Mixed signal VLSI.

EXPERIENCE

LLM Localization Hardware Accelerator **Nov 2025 - Present**
AI EDGE - The Ohio State University, the National Science Foundation *Columbus, OH, USA*

- Spearheading the design of a hardware accelerator focused on localizing Large Language Models (LLMs) for efficient edge deployment.
- The architecture specifically targets accelerating key NLP operations: high-speed encoding, decoding, compression, and decompression of LLM data

Embedded Systems Research and Development Intern **Jan. 2024 – Jun 2024**
Department of Aeronautics, Indian Institute of Science Bangalore *Bangalore, India*

- Engineered and built a specialized drone thrust bench using an Arduino microcontroller to control the hardware and capture real-time propulsion data from load cells and a Hall sensor. The project source code and report can be found in my GitHub page.
- Developed a Python-based graphical user interface (GUI) to retrieve and display live sensor data. The GUI includes features for throttle control, real-time data display of thrust, torque, current, and voltage, and a safety HALT function to shut down the system.
- Validated the system's reliability and accuracy by comparing its measurements with readings from a commercial wattmeter and a standard Benchmark thrust bench, demonstrating a strong correlation between the data sets.

IoT (Internet of Things) Intern **Jun 2022 – Aug 2022**
Telibrahma Software Services LLP *Bangalore, India*

- Developed an environmental monitoring system for poultry management, utilizing an ESP32 microcontroller and a BME680 environmental sensor to collect readings of a range of factors affecting poultry health.
- Engineered an embedded system that authenticates with a Firebase, sending real-time sensor data for temperature, humidity, and pressure.
- Implemented a secure data pipeline that stores environmental readings in a Firebase Realtime Database, with security rules restricting access to a unique user ID. The project source code and report can be found in my GitHub page.

PROJECTS

American Sign Language Interpreter Glove: Designed and built an ASL Interpreter Glove, developing firmware for an ESP32 microcontroller to capture real-time sensor data from flex sensors and a gyroscope. Created a custom PCB for the device and implemented a real-time data pipeline that transmitted sensor readings over Wi-Fi to a cloud database for processing. The outcome is the conversion of signs to the English alphabet and words. The project source code and report can be found in my GitHub page.
VLSI Perceptron Accelerator: Designed the complete schematic and layout of the 32-word on-chip SRAM memory unit, the core storage element for a pipelined perceptron. Designed and integrated critical sub-blocks, including the 6T SRAM cell, high-speed Kogge Stone Adder and a multiplier to ensure efficient, synchronized read/write operations. The Source code can be found in my GitHub page.

PUBLICATION

LORV: Latency-Optimized Routing for NoC with Virtual Channels in 2D Mesh: Proposed an adaptive routing algorithm for efficient data transfer in Network on Chips. This project has been converted into a paper and has been published by IEEE Computer Society. The paper can be found here.

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Concepts: Routing, Computer and Embedded System Arch., VLSI Design, Design Verification, Embedded Firmware, PCB Design, Micro controllers, Digital Electronics, Computer Hardware, Internet of Things, Mixed signal VLSI.

EXPERIENCE

Researcher - LLM Localization Hardware Accelerator **Nov 2025 - Present**
AI EDGE The Ohio State University, the National Science Foundation *Columbus, OH, USA*

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- Implemented a secure data pipeline that stores environmental readings in a Firebase Realtime Database, with security rules restricting access to a unique user ID. The project source code and report can be found in my GitHub page.

PROJECTS

Discrete-Event Simulation of Network Queueing Systems Developed a discrete-event simulation to evaluate network traffic performance, demonstrating that standard M/M/1 theoretical models fail to accurately predict packet delays in complex, parallel routing topologies due to traffic correlation and burstiness. The project source code and report can be found in my GitHub page.
VLSI Perceptron Accelerator: Designed the complete schematic and layout of the 32-word on-chip SRAM memory unit, the core storage element for a pipelined perceptron. Designed and integrated critical sub-blocks, including the 6T SRAM cell, high-speed Kogge Stone Adder and a multiplier to ensure efficient, synchronized read/write operations. The Source code can be found in my GitHub page.

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EDUCATION

The Ohio State University: ***Master's in Electrical and Computer Engineering*** **(GPA: 3.5/4)** **Jun 2026**

PES University: ***Bachelor's in Electrical and Eommunication Engineering*** **(GPA: 3.5/4)** **Jun 2024**

SKILLS

- **Programming & Tools:** C, Python, Verilog, SystemVerilog, VHDL, FPGA, Xilinx, Quartus MATLAB, SPICE, KiCAD, Git, Shell, Cadence, Excel, LATEX, Assembly, Wireshark, GNS3, Raspbian, SQL, RTOS, Sensors, Communication protocols, Mixed Signal Design, FPGA Verification, Digital Design
- **Concepts:** Routing, Computer and Embedded System Arch., VLSI Design, Design Verification, Embedded Firmware, PCB Design, Micro controllers, Digital Electronics, Computer Hardware, Internet of Things, Mixed signal VLSI, Digital Design Automation

EXPERIENCE

AI EDGE - the National Science Foundation **Nov 2025 - Present**
Researcher - LLM Localization Hardware Accelerator *Columbus, OH, USA*

- Designed and implemented a complete FPGA Huffman compression pipeline in synthesizable SystemVerilog, comprising a runtime-configurable encoder with 256-symbol BRAM lookup table and canonical decoder with three configurable lookup tables, connected via a valid/ready handshake interface.
- Built a 64-bit MSB-first accumulator datapath with 32-bit streaming output, achieving 1 symbol/cycle throughput at 146 MHz post place-and-route on Xilinx FPGA; verified end-to-end round-trip correctness across three gradient distribution scenarios in Vivado behavioural simulation.
- Demonstrated 58.5% reduction in gradient transfer time for a GPT-2 model (248 MB, bfloat16) over a 1 Gbps link under sparse gradient conditio ns (2.41× compression), quantifying hardware compression impact on distributed LLM training communication overhead.

Department of Aeronautics, Indian Institute of Science Bangalore **Jan 2024 - Jun 2024**
Embedded Systems Research and Development Intern *Bangalore, India*

- **Engineered a specialized drone thrust bench** for **Rs 5,000**, replicating features of a **Rs 1.5L** commercial system (96% cost reduction).The project source code can be found in my GitHub page.
- **Developed a Python GUI** with YAML-based constraint input automation for real-time tracking of thrust, torque, and electrical metrics.
- Validated the system's reliability and accuracy by comparing its measurements with readings from a commercial wattmeter and a standard Benchmark thrust bench, demonstrating a strong correlation between the data sets. Integrated safety HALT functions and digital workflows to streamline propulsion testing.

Telibrahma Software Services LLP **Jun 2022 - Aug 2022**
IoT (Internet of Things) Intern *Bangalore, India*

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PROJECTS

ASL Interpreter Glove (GitHub Page)**:** Co-developed a wearable American Sign Language translation system using ESP32 microcontrollers, flex se nsors, and IMUs to translate static signs into English text in real-time. Implemented a Random-Forest Machine Learning model that achieved **93.3%** accuracy on a demonstration subset by applying temporal smoothing algorithms to stabilize inference. Architected a wireless IoT pipeline using Fireb ase to synchronize dual-glove data without physical tethering and designed a custom PCB to integrate components into a compact form factor, optimi zing trace widths for signal integrity

VLSI Perceptron Accelerator (GitHub page): Designed a full-stack neural network accelerator from 6T SRAM cell level to architectural integr ation, featuring a custom 32-word SRAM for weight storage, Kogge-Stone Adder for reduced logic depth, and an 8-bit multiplier with shifting suppo rt. Engineered to handle 32-bit data streams with a ReLU activation function for scalable neural network inference.

Discrete Breadboard Graphics Card: Designed a microcontroller-free VGA card using discrete ***74LS logic*** and a V-Model flow, engineering custo m timing circuits to adapt a ***10 MHz*** oscillator for **100x75** resolution. Integrated an ***AT28C256 EEPROM*** for image storage and designed a resistor-network DAC to generate ***64-color*** analog RGB video signals.

PUBLICATION

LORV: Latency-Optimized Routing for NoC with Virtual Channels in 2D Mesh: *Developed an adaptive routing algorithm for 2D Mesh NoCs that reduced latency by **50%** and hop counts by **4.8%**. Integrated Virtual Channels to eliminate deadlocks and optimize data transfer for large-scale packets. The paper can be found here and the source code can be found in my GitHub Page.*